



Project: Assessment and Calibration of Polarimetric P-band BIOMASS SAR

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CCRS

Natural Resource of Canada

Acknowledgment

Clement Albinet and Björn Rommen (ESA) for the helpful information on Biomass, and ESA for the Biomass Data



OUTLINE



- Lessons learned from PALSAR & PALSAR2 calibration using CR and rainforests.
- Assessment of BIOMASS antenna cross-talks using JAXA CRs in the Amazonian rainforests, **at low Faraday rotation conditions**
- Assessment of BIOMASS calibration using the ESA transponder in Australia.
- Future tasks

PALSAR3 Antenna Xtalk Measurement Lessons learned from PALSAR & PALSAR2

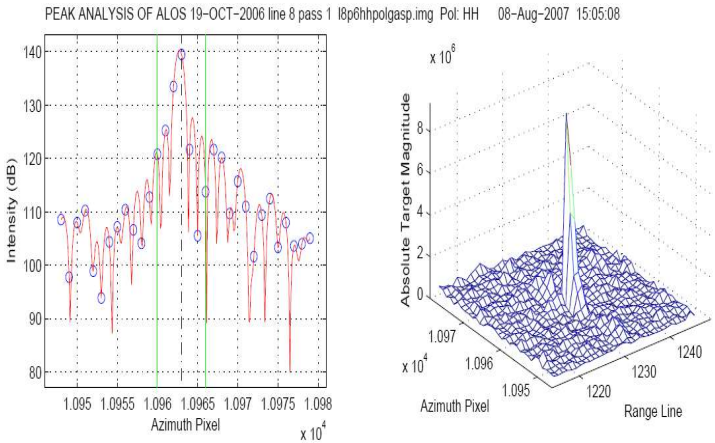
- Big confusion at CVST6 (Mar.07) :
  - Antenna Xtalk or Faraday rotation?!!!
  - ➡ Mis-leading conclusions: PALSAR antenna Xtalk (-23dB) at PolinSAR 2007
- ☛ Solved at the ALOS-CVST7 (19 June 2008)
  - **Pure** Xtalk measurement => Calibration parameters should be conducted at **low Faraday**
  - **R. Touzi and M. Shimada, "Calibration of Polarimetric PALSAR", *IEEE TGRS*, ALOS special issue, Vol. 47, No.12, pp. 3911-3913, Dec. 2009**
- ➡ Demonstration using the Sweden 5m CR

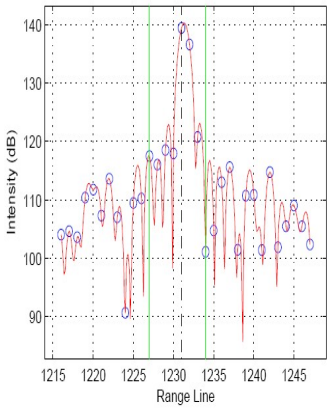
PALSAR Assessment using the Sweden CR



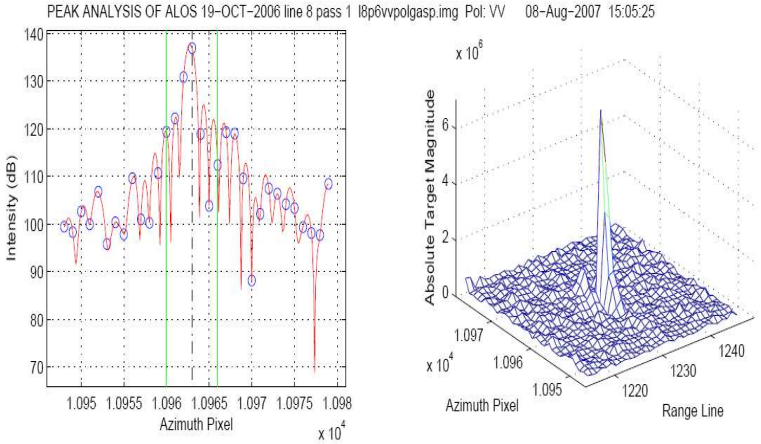
- 5m-height trihedral with a cross section of 47.2 dBm.
- Aknowledgment: Leif Eriksson and G. Sabdberg, Chalmers University of Technology

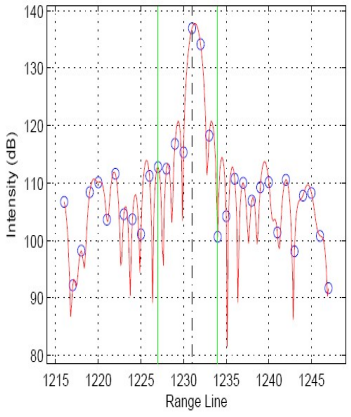
Day acquisition (5 Oct. 06)
HH and VV





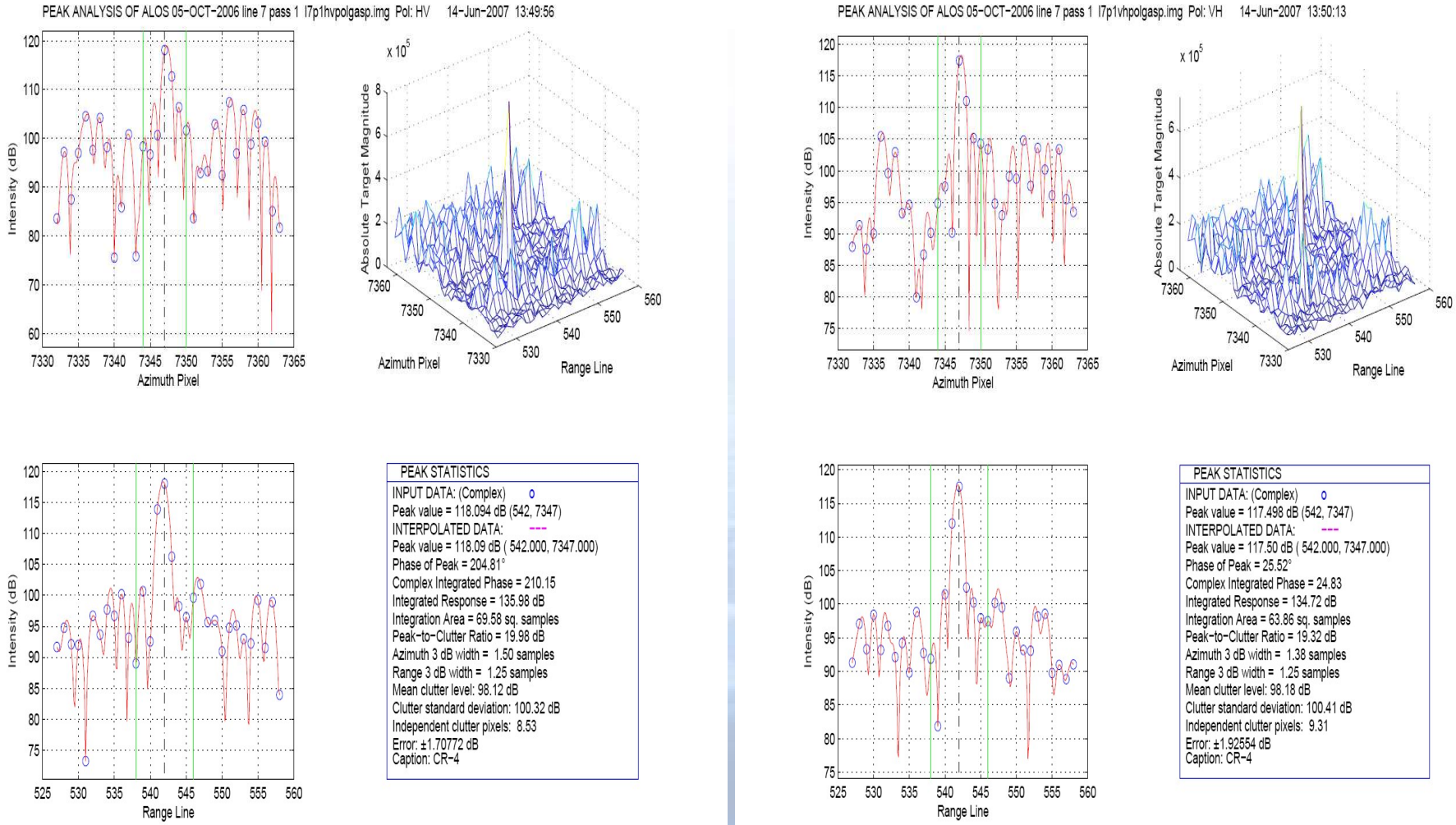
PEAK STATISTICS
INPUT DATA: (Complex) 0
Peak value = 139.443 dB (1231, 10963)
INTERPOLATED DATA: ---
Peak value = 140.23 dB (1230.750, 10963.375)
Phase of Peak = 21.37°
Complex Integrated Phase = 23.00
Integrated Response = 158.67 dB
Integration Area = 57.58 sq. samples
Peak-to-Clutter Ratio = 32.62 dB
Azimuth 3 dB width = 1.38 samples
Range 3 dB width = 1.12 samples
Mean clutter level: 108.57 dB
Clutter standard deviation: 109.25 dB
Independent clutter pixels: 10.34
Error: ±0.07683 dB
Caption: CR-1





PEAK STATISTICS
INPUT DATA: (Complex) 0
Peak value = 136.967 dB (1231, 10963)
INTERPOLATED DATA: ---
Peak value = 137.73 dB (1230.750, 10963.375)
Phase of Peak = 43.37°
Complex Integrated Phase = 44.94
Integrated Response = 156.19 dB
Integration Area = 57.58 sq. samples
Peak-to-Clutter Ratio = 33.88 dB
Azimuth 3 dB width = 1.38 samples
Range 3 dB width = 1.12 samples
Mean clutter level: 104.82 dB
Clutter standard deviation: 105.42 dB
Independent clutter pixels: 10.34
Error: ±0.05654 dB
Caption: CR-1

Day acquisition (5 Oct. 06)
HV and VH: **Significant** return
➪**Mis-leading** conclusions on PALSAR antenna (PolinSAR 2007)





Faraday rotation or Antenna Cross Talk???!!!



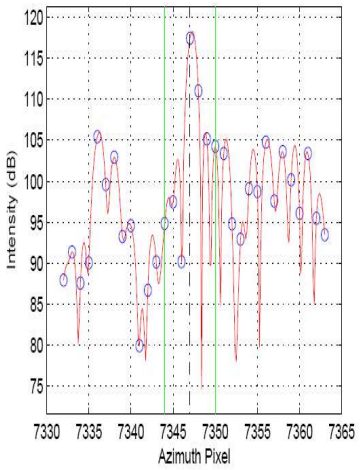
- After JAXA calibration:

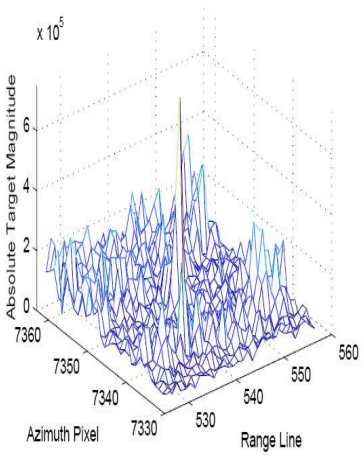
$$M_{hh} = S_{hh}\cos^2\Omega - S_{vv}\sin^2\Omega$$
$$M_{hv} = f_1\left[S_{hv} + (S_{hh} + S_{vv})\sin\Omega\cos\Omega\right]$$
$$M_{vh} = f_1\left[S_{hv} - (S_{hh} + S_{vv})\sin\Omega\cos\Omega\right]$$
$$M_{vv} = f_1^2\left[S_{vv}\cos^2\Omega - S_{hh}\sin^2\Omega\right]$$

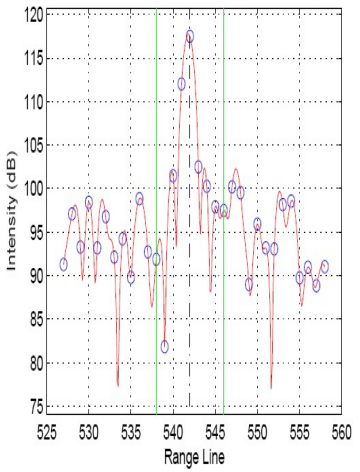
- Faraday rotation => HV+VH=0 for a CR

Sweeden Calibrated Data HV + VH
Day acquisition (5 Oct. 06)
HV+VH Low => HV peak due to Farday rotation

HV

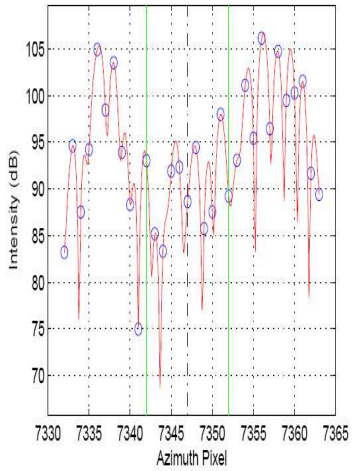


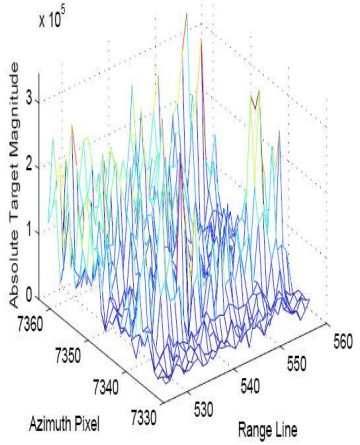


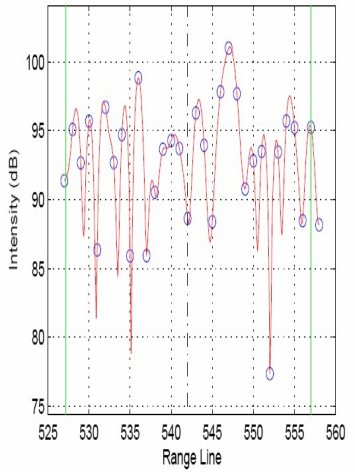


PEAK STATISTICS
INPUT DATA: (Complex) o
Peak value = 117.498 dB (542, 7347)
INTERPOLATED DATA: ---
Peak value = 117.50 dB (542.000, 7347.000)
Phase of Peak = 25.52°
Complex Integrated Phase = 24.83
Integrated Response = 134.72 dB
Integration Area = 63.86 sq. samples
Peak-to-Clutter Ratio = 19.32 dB
Azimuth 3 dB width = 1.38 samples
Range 3 dB width = 1.25 samples
Mean clutter level: 98.18 dB
Clutter standard deviation: 100.41 dB
Independent clutter pixels: 9.31
Error: ±1.92554 dB
Caption: CR-4

HV+VH







PEAK STATISTICS
INPUT DATA: (Complex) o
Peak value = 88.644 dB (542, 7347)
INTERPOLATED DATA: ---
Peak value = 88.64 dB (542.000, 7347.000)
Phase of Peak = 194.91°
Complex Integrated Phase = 340.44
Integrated Response = 0.00 dB
Integration Area = 364.00 sq. samples
Peak-to-Clutter Ratio = -8.96 dB
Azimuth 3 dB width = 16.88 samples
Range 3 dB width = 1.88 samples
Mean clutter level: 97.60 dB
Clutter standard deviation: 100.16 dB
Independent clutter pixels: 0.51
Error: ±13.85272 dB
Caption: CR-4



Faraday rotation Measurement



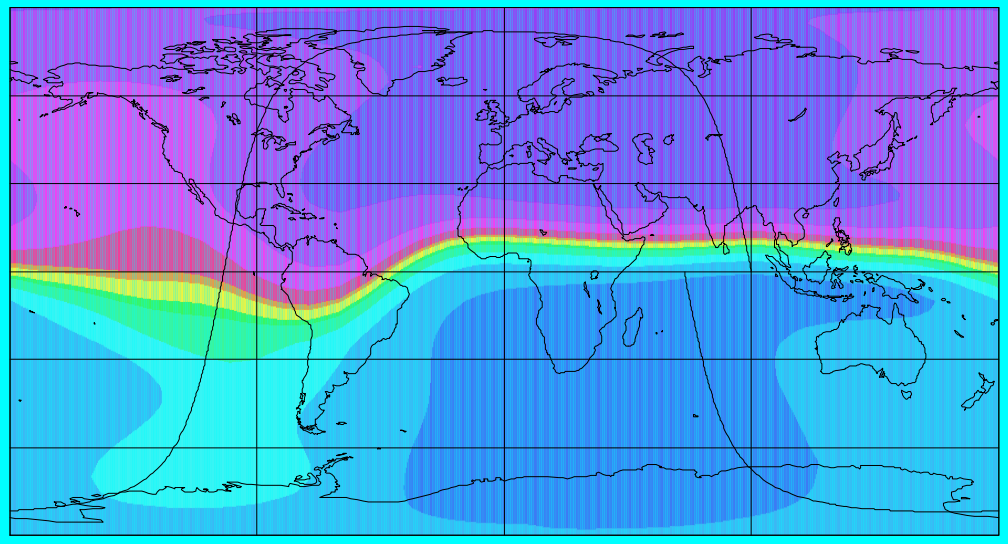
- Sweden site Oct 5, Significant Faraday rotation:
  - HV+VH= $2*S_{HV} \cong 0$
  - HV & VH => Xtalk + Faraday rotation contamination
- CR deployed at the Amazonian rainforest at low Faraday
- Measurement of pure Antenna Xtalks.
- PALSAR Distortion matrix
  - $|\Delta_3|$= -37.98 dB, $|\Delta_4|$=-42.60 dB

  $|f_2/f_1|$=1.004, $|f_1|$=0.99, $|f_2|$=0.985

PALSAR calibration parameters should be conducted at Low Faraday rotation => Amazonian rainforests

R. Touzi and M. Shimada, "Calibration of Polarimetric PALSAR", *IEEE TGRS*, ALOS special issue, Vol. 47, No.12, pp. 3911-3913, Dec. 2009

Predictions of one way FR under high sunspot activity for 12:00 GMT, April and 35 degree incidence angle



- Typical values of absolute two-way FR (at L-band) in most of the Amazon between 0-10º
- Wright and Quegan, Faraday rotation effects on L-band spaceborne SAR data. *IEEE TGRS*, Vol. 41, No. 12, pp. 2735-2744.

CR deployed in the Amazonian rainforest for PALSAR2 (Shimada & Isoguchi ☺:)

Pol Mode: Antenna Pattern

FP高分解能[6m]モード観測性能

ビームNo		1	2	3	4	5	6	7	
オフナディア角[deg]		17.8	21.5	25	28	30.4	32.7	34.9	
TRモジュール	送信	180	180	180	180	180	180	180	
使用台数	受信	90×2	90×2	90×2	90×2	90×2	90×2	90×2	デュア
入射角範囲[deg]	Near	17.3	21.6	25.6	29.4	32.3	35	37.6	
	Far	21.8	25.9	29.7	32.5	35.2	37.8	40.2	
観測幅[km]		50	50	50	40	40	40	40	
分解能[m]	Rg	9.5	7.9	6.8	6.2	5.7	5.3	5.1	
	Az	4.3	4.3	4.3	4.3	4.3	4.3	4.3	
NESZ[dB]		-30.8	-29.8	-29.3	-28.2	-27.4	-27.2	-26.2	
S/A[dB]	Rg(ライク偏波)	44.5	30.6	37.4	37.7	34.4	33.2	32.8	+10dE
	Rg(クロス偏波)	35.3	30.6	27.6	27.9	24.4	24.5	24	+10dE
	Az	23	22.5	24.6	23	22.7	24.5	23.1	
データレート[Mbps]		510	570	720	600	620	730	700	ダウン
観測偏波		FP							

※1:マーキング部は性能規定値。

FP7

FP3

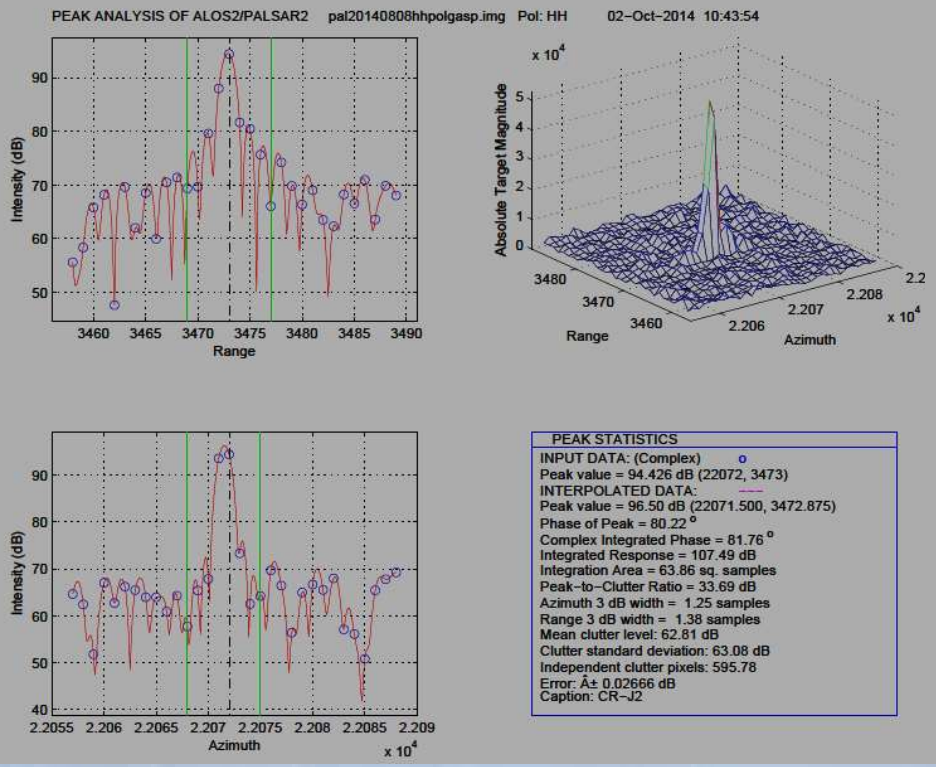
FP3

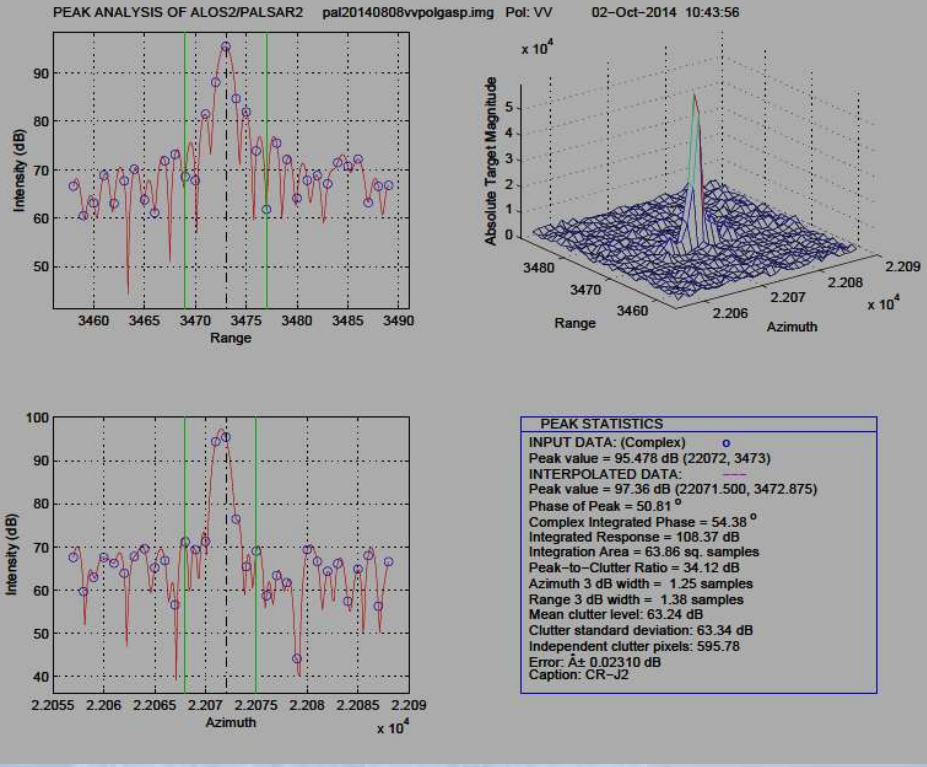
Rio Branco CR site



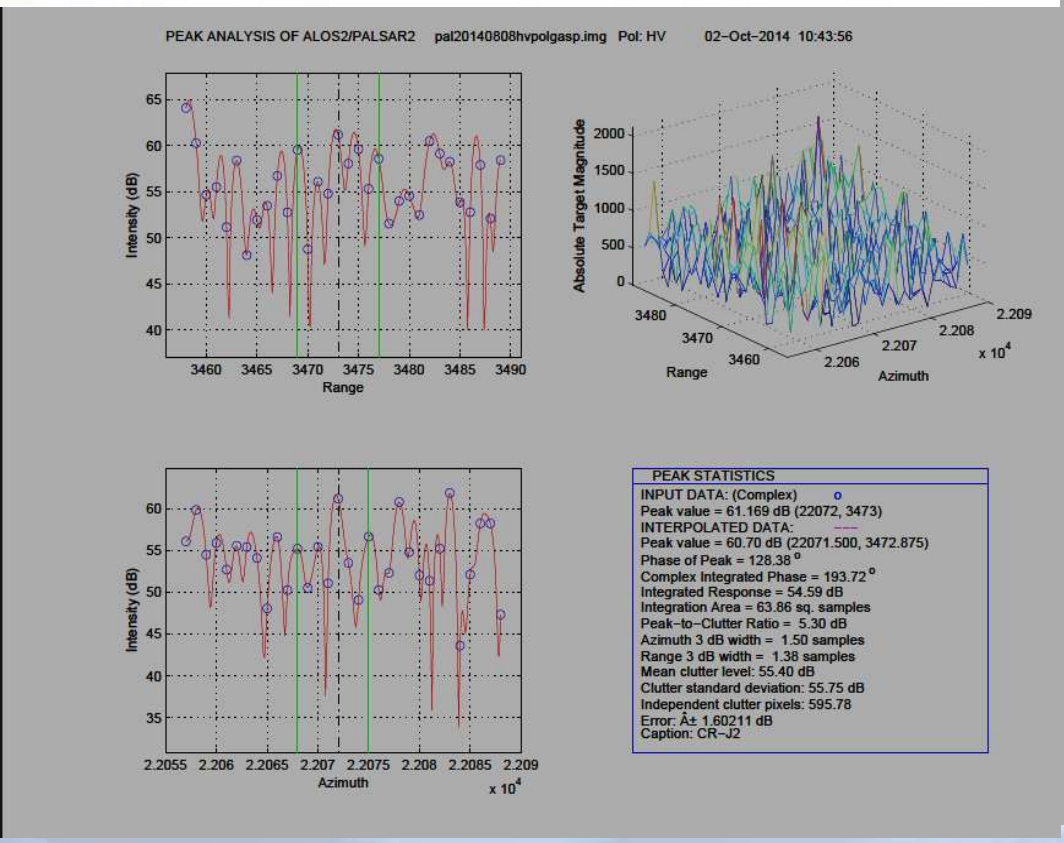
Name	File name	Date	Latitude	Longitude	Latitude	Longitude	Grid Northing (m)	Grid Easting (m)	Ell-Height (m)
J1 Corner	SARD092200029296-00012.tgz	13-Aug-14	9° 28'31.05585"S	68° 22'24.81009"W	-9.475294°	-68.373558°	8952538.148	568763.774	205.88
	SARD092200029296-00012.tgz	27-Aug-14							
J2 Corner	SARD091700028510-00011.tgz	8-Aug-14	9° 38'35.51883"S	68° 02'04.76752"W	-9.643200°	-68.034658°	8933886.719	605915.324	178.364
	SARD092200028510-00011.tgz	8-Aug-14							
J3 Corner			9° 26'43.64484"S	67° 43'41.92800"W	-9.445456°	-67.728313°	8955644.453	639611.879	196.400
J4 Corner	SARD092200036811-00006.tgz	5-Sep-14	9° 06'52.16591"S	67° 19'15.79346"W	-9.114492°	-67.321053°	8992061.712	684506.539	149.637
J5 Corner			9° 15'00.58400"S	67° 01'50.62842"W	-9.250161°	-67.030730°	8976892.595	716340.219	158.959

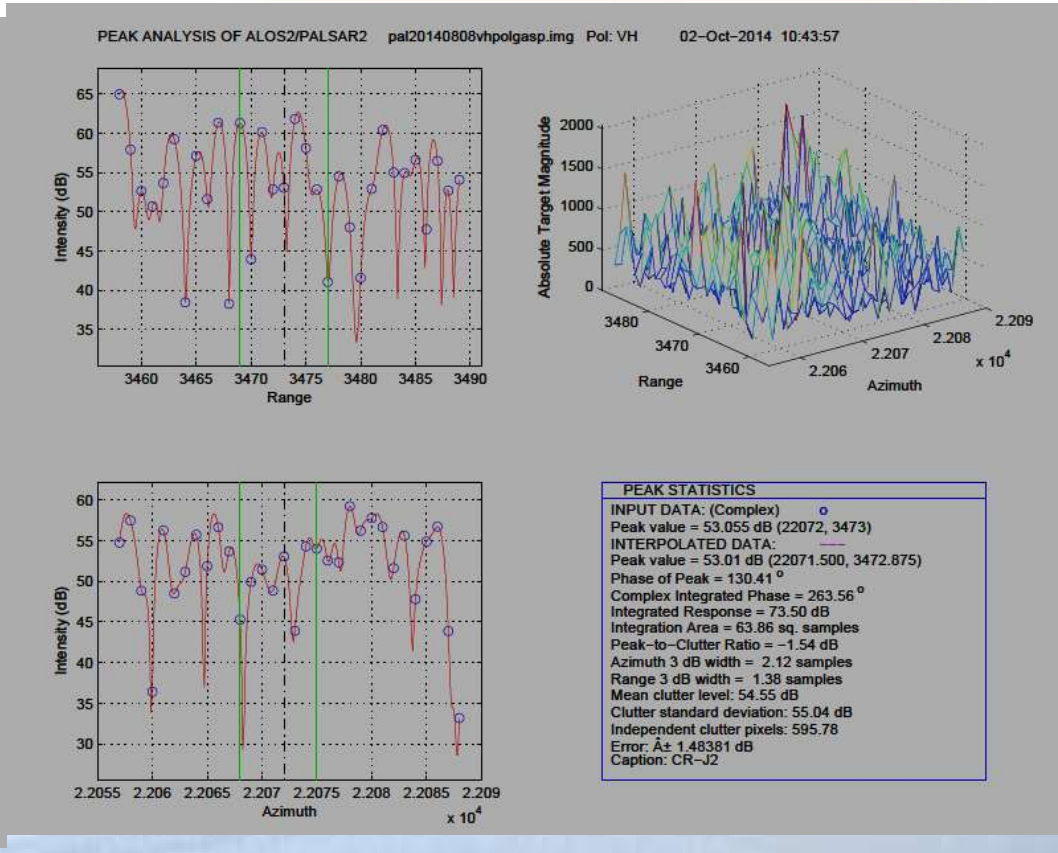
PALSAR2 parameters using uncalibrated Amazonian data (FP3, 8 Aug 2014) HH and VV





PALSAR2 parameters using uncalibrated Amazonian data (FP3, 8 Aug 2014) HV and VH





☛ No CR return at HV and VH => Very low Xtalk

PALSAR2 Antenna Xtalk (FP3-to-FP7) using uncalibrated (antG corrected) Amazonian data

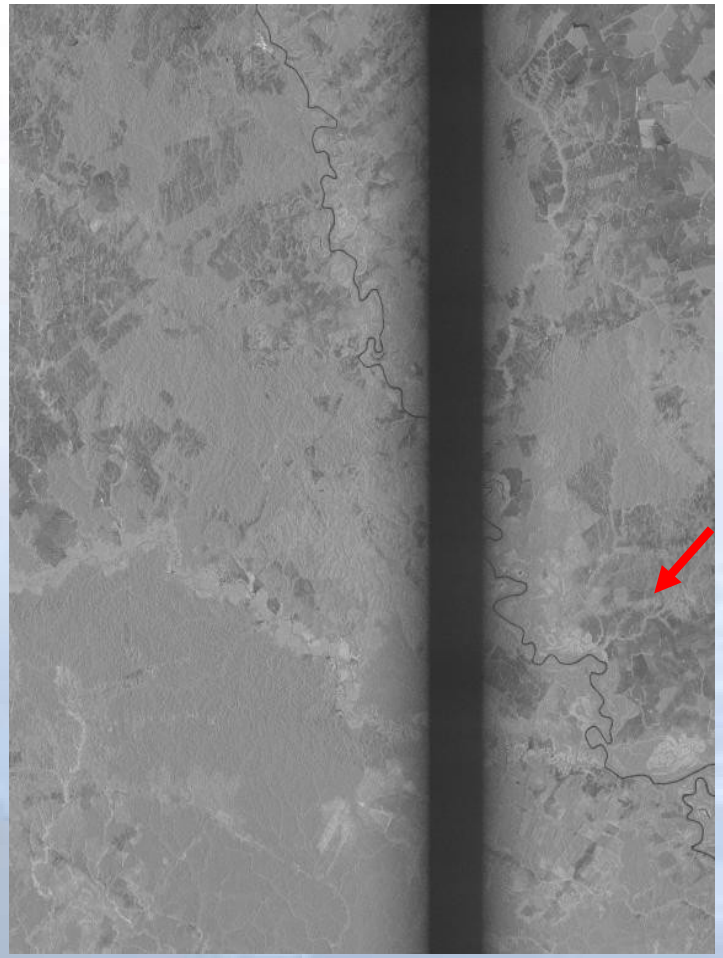


- **FP3, 8 aug 2014**
  - **$|\Delta_3|$= -40.38 dB, $|\Delta_4|$=-45.31 dB**
- **FP4, 22 aug 2014**
  - **$|\Delta_3|$= -44.51, $|\Delta_4|$= = -41.8**
- **FP5, 5 Sep2014**
  - **$|\Delta_3|$= -40.68 dB, $|\Delta_4|$=-41.20 dB**

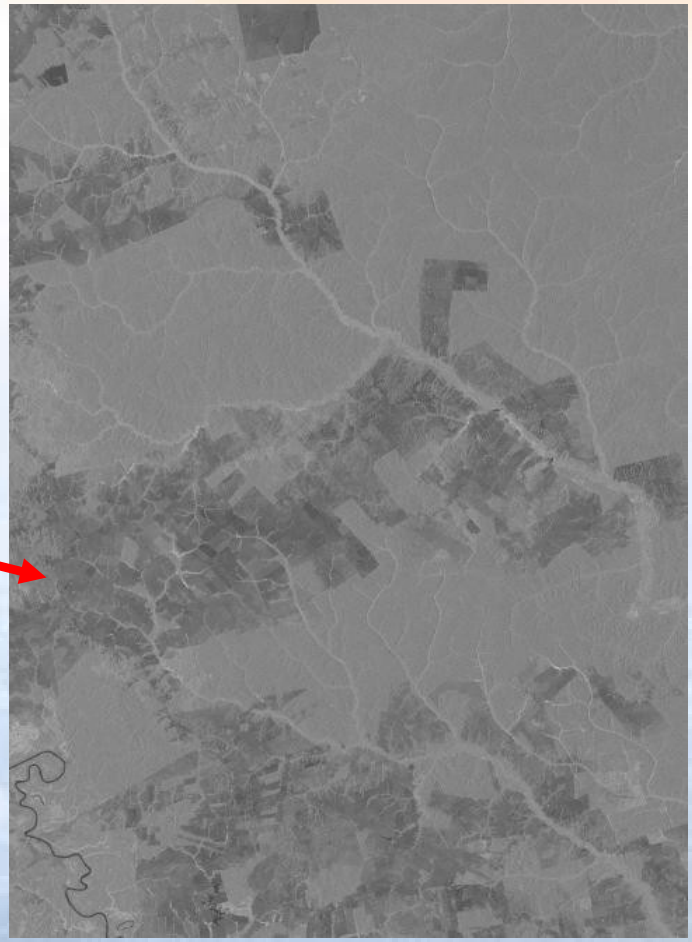
Excellent antenna design =>Xtalk lower than -40dB

R. Touzi, M. Shimada, and T. Motohka, "Calibration and Validation of Polarimetric ALOS2", Rem. Sens., Special issue on ALOS2 Calibration, Vol. 14, Num. 2452, pp. 1-13, May 2022

PALSAR 3 (HQB06) 1 March 2025



ALOS40287000250301-HBQPRA0606 (30°-34°)



ALOS40287000250301-HBQPRA0607 (34°-38°)

Assessment of PALSAR3 Distortion matrix using the PALSAR3 image corrected for the Antenna Gain

- Freeman's symmetrization Model:

$$\begin{bmatrix} V_{hh} & V_{vh} \\ V_{hv} & V_{vv} \end{bmatrix} = Ae^{j\phi} \begin{bmatrix} 1 & \Delta_3 \\ \Delta_4 & f_1 \end{bmatrix}^T \begin{bmatrix} S_{hh} & S_{vh} \\ S_{hv} & S_{vv} \end{bmatrix} \begin{bmatrix} 1 & \Delta_3 \\ \Delta_4 & f_2 \end{bmatrix}$$

$$\Delta_3 = \frac{|f_2|}{|f_1|} e^{-j\phi_r} (\delta_1 + \delta_3 f_1 / f_2)/2$$

- Combination with Van Zyl model:

$$\Delta_4 = (\delta_2 + \delta_4)/2$$

$$\begin{bmatrix} V_{hh} & V_{vh} \\ V_{hv} & V_{vv} \end{bmatrix} = Ae^{j\phi} \begin{bmatrix} 1 & \Delta_3 \\ \Delta_4 & f_1 \end{bmatrix}^T \begin{bmatrix} S_{hh} & S_{vh} \\ S_{hv} & S_{vv} \end{bmatrix} \begin{bmatrix} 1 & \Delta_3 \\ \Delta_4 & f_2 \end{bmatrix}$$

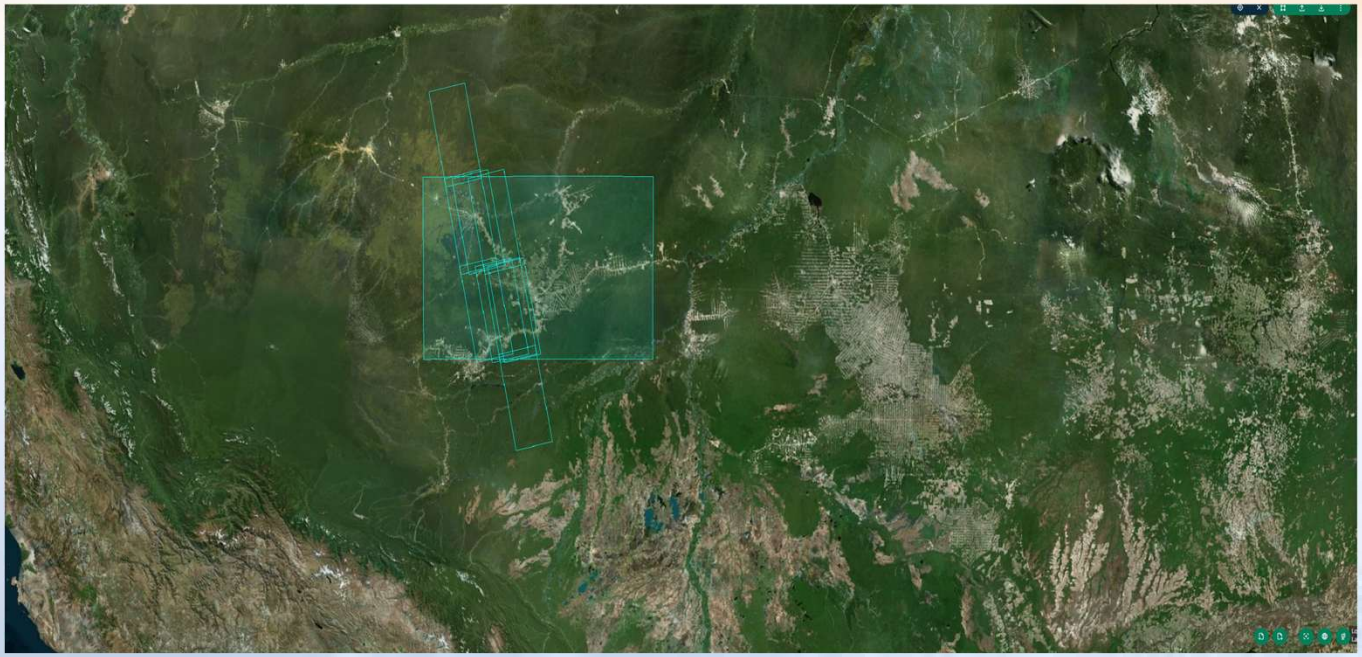
- **Extended Freeman-VanZyl Calibration** Model (Touzi & Shimada TGRS 2009):
- **PALSAR3 Distortion matrix (34°)**
  - **$|\Delta_3|$= -39.28 dB, $|\Delta_4|$=-43.07 dB, $|f_1|$=1.008, $|f_2|$=0.985**
- R. Touzi, M. Shimada, and T. Motohka, "Calibration and Validation of Polarimetric PALSAR3", ALOS-PI workshop, Tokyo, Oct. 2025

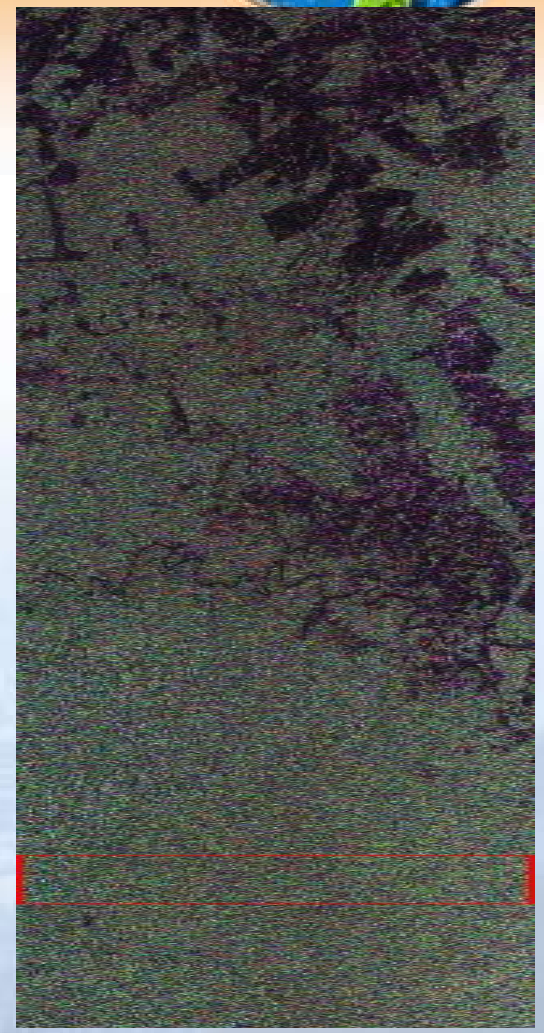
Assement of BIOMASS: RioBranco CR

CR	Latitude (deg.)	Longitude (deg.)	Height (km)	Azimuth (deg.)	Elevation (deg.)	orbit	size(m)	Path	Beam
J1 Corner	−9.475294°	−68.373558°	0.205880	258.15	51.22	A	3.0	29	FP6−7
J2 Corner	−9.643200°	−68.034658°	0.178364	258.15	62.63	A	3.0	28	FP6−3
J3 Corner	−9.445456°	−67.728313°	0.196400	258.15	59.63	A	3.0	28	FP6−4
J4 Corner	−9.114492°	−67.321053°	0.149637	258.15	55.74	A	3.0	28	FP6−5
J5 Corner	−9.250161°	−67.030730°	0.158959	258.15	53.75	A	3.0	28	FP6−6



Assessment of BIOMASS Antenna Xtalk using JAXA CRs





- BIO_S1_SCS__1S_20250710T102742_20250710T102803
- BIO_S3_SCS__1S_20251104T102440_20251104T102501
- BIO_S3_SCS__1S_20251104T102459_20251104T102520

3m CR not detected by the P-band BIOMASS SAR



Assessment of BIOMASS Calibration using the Transponder



- **Uncalibrated** data are **required** for the assessment of BIOMASS antenna distortion matrix.
- **Calibrated** BIOMASS data provided.
- Assessment of BIOMASS calibration

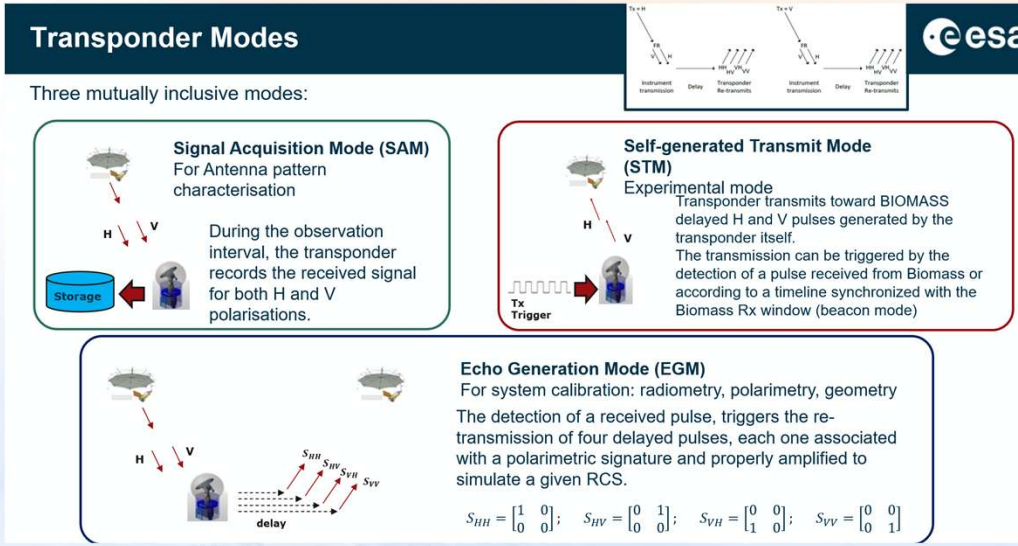
Assessment of BIOMASS calibration using the Transponder

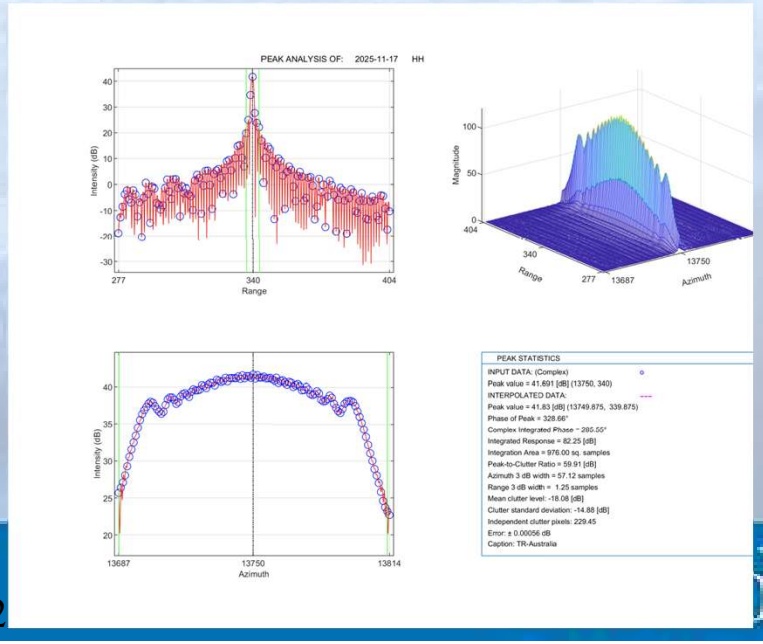
- Transponder: Echo Mode
- Analysis of transponder

  response => Reesidual

  Antenna Xtalks .

$$\begin{bmatrix} M_{hh} \\ M_{hv} \\ M_{vh} \\ M_{vv} \end{bmatrix} = \begin{bmatrix} S^{\Omega}_{hh} + \delta_4 S^{\Omega}_{hv} + \delta_2 S^{\Omega}_{vh} \\ \delta'_3 S^{\Omega}_{hh} + S^{\Omega}_{hv} + \delta_2 S^{\Omega}_{vv} \\ \delta'_1 S^{\Omega}_{hh} + S^{\Omega}_{vh} + \delta_4 S^{\Omega}_{vv} \\ \delta'_1 S^{\Omega}_{hv} + \delta'_3 S^{\Omega}_{vh} + S^{\Omega}_{vv} \end{bmatrix}$$





Assessment of Calibrated Biomass images

- 31 Jan. 2026:
  - δ_1=-37.89 δ_2= - 28.86 δ_3=-27.10 δ_4=- 27.10 dB
- 13 Jan. 2026:
  - δ_1=-36.00 δ_2= =- 29.53 δ_3=-30.51 δ_4=- 34.21 dB
- 17 Nov. 2025:
  - δ_1=-28.34 δ_2=-34.64 δ_3=-34.59 δ_4=-37.70 dB

- **Residual** Xtalks: **Large variations (up to 10dB)** due to the **different** date **Faraday rotation** contaminations.
- Xtalks **not** originally **well calibrated**
- Future work: uncalibrated data for transponder Cal-Val.
- New method using BIOMASS data on the amazonian forest at low Faraday conditions

BIOMASS Excellent NESZ

- More accurate calibration required for low HV applications
- Key applications (Low HV):
  - Subsurface permafrost monitoring in the Arctic and Subarctic regions
  - Detection & mapping of desert water structures buried beneath the Sahara Desert (Paillou et al.)

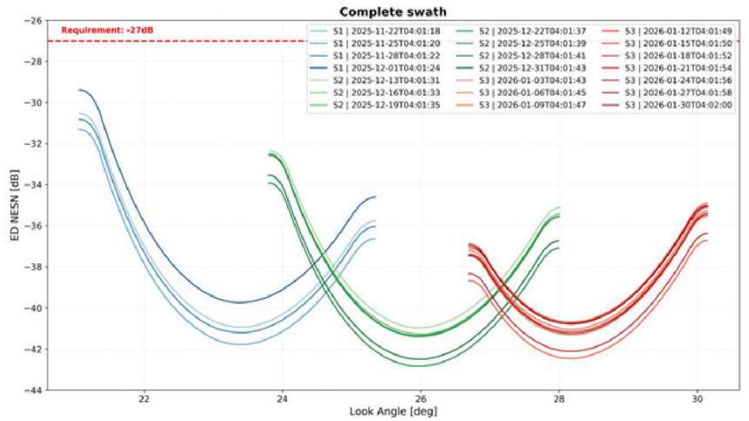
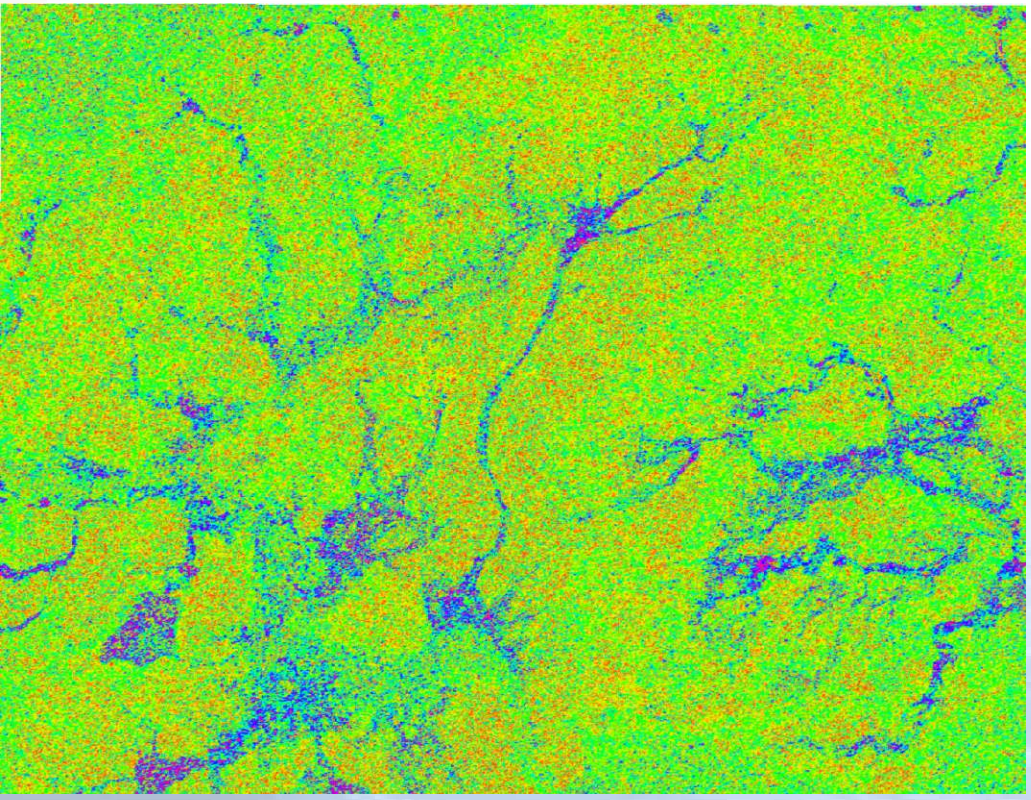


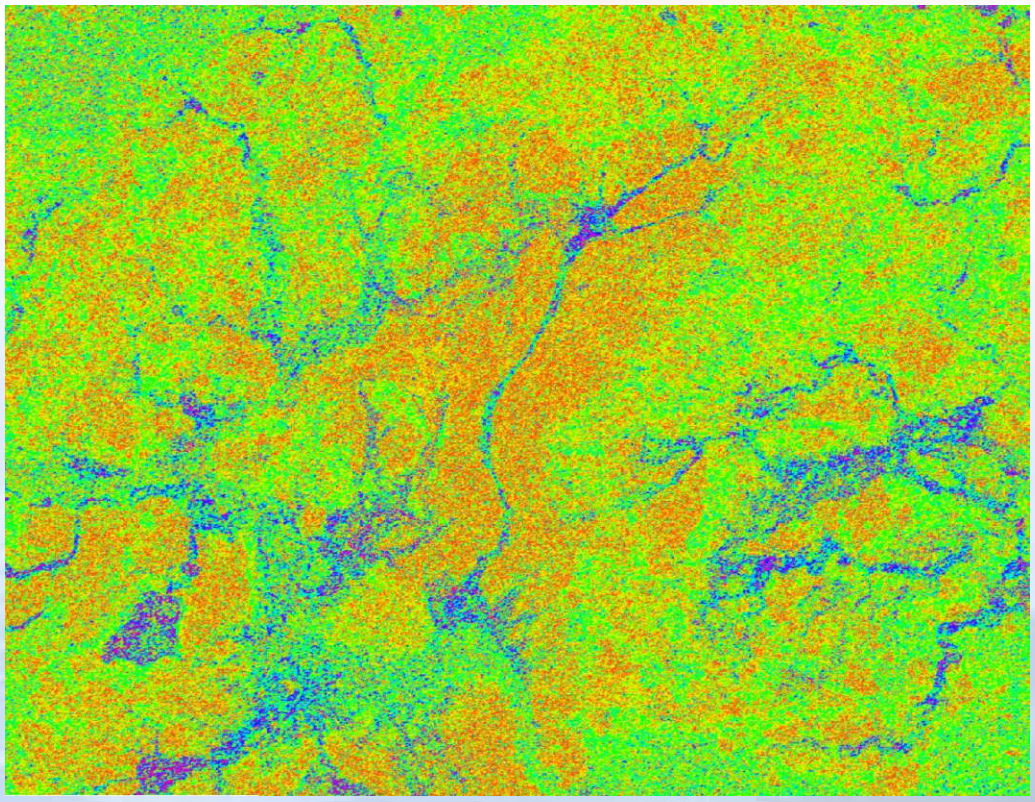
Figure 8 NESN profiles estimated from 21 data takes acquired over swaths 1, 2 and 3. All NESN profiles are compliant to the threshold of -27 dB.

A.B.C. Silva et al., Calibration and Validation Results for Biomass Operational Phase, EUSAR 2026

Example: PALSAR3 FP6-4 miscellenous calibration (-33dB) PALSAR3 ReCalibrated to Fully Exploit the Excellent NESZ (-38dB for enhanced subsurface permafrost mapping & monitoring

Touzi Decomposition => Scattering type phase



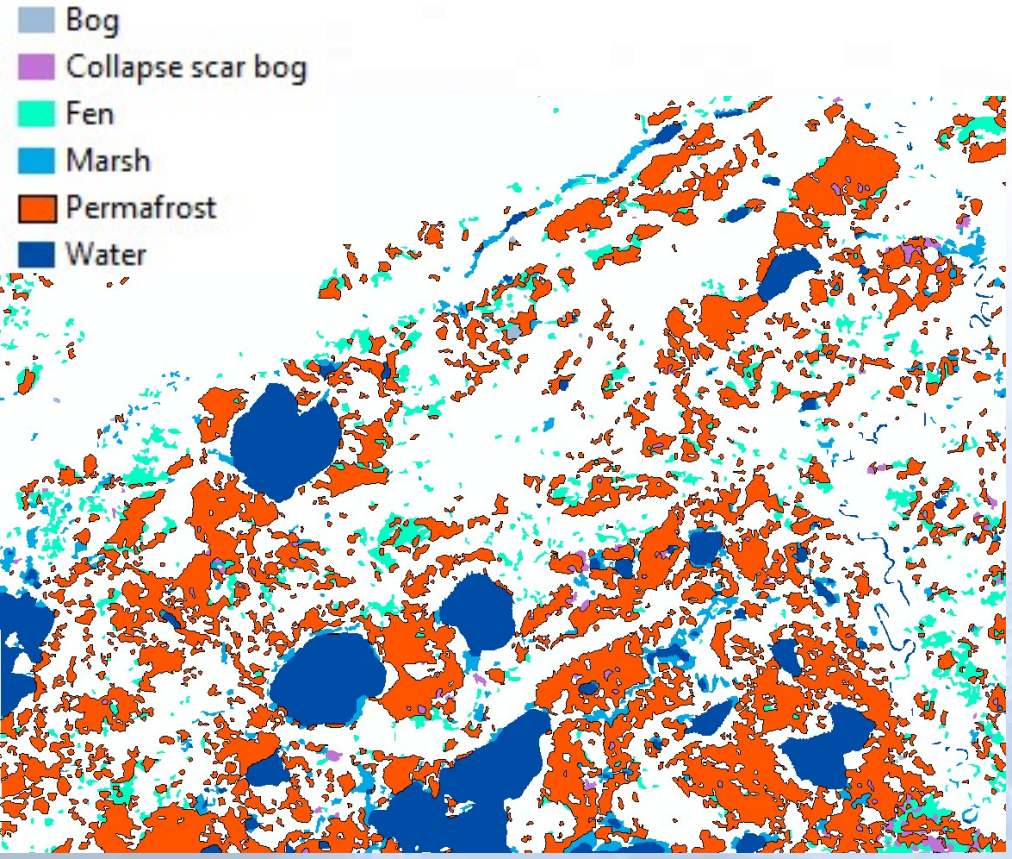


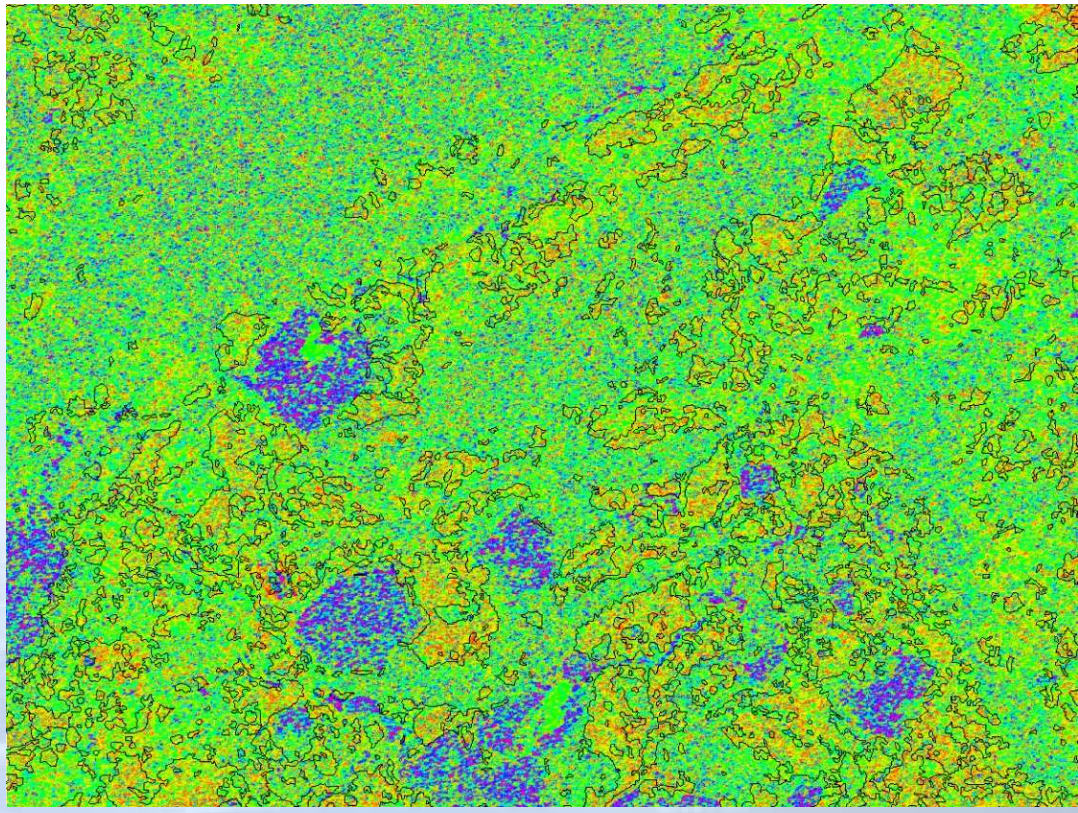
Original ALOS2 residual error -33dB

Re-calibrated residual error -45dB

- Permafrost **detection** => **Low** NESZ for **accurate** **HV measure & ICTD**
- Re-Calibration => fully exploting the excellent ALOS2 NESZ (-38dB)

Polarimetric PALSAR2: FP6-4 Aug. 2014 Lidar-Landsat Permafrost Classification versus ALOS2 Touzi Scattering Phase (permafrost in Pink)





Excellent ALOS2 NESZ (-38dB) => Polarimetric ALOS2 Very promising for subsurface permafrost Mapping (orange in the Phase) and Monitoring